Status Note: EAE/MC to SC for Banach Space Operators

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Status. `literature_already_answered`. This is a literature-status packet, not a new proof or counterexample.

Source Question

The source paper is S. ter Horst, M. Messerschmidt, A. C. M. Ran, M. Roelands, and M. Wortel, *Equivalence after extension and Schur coupling coincide for inessential operators*, arXiv:1706.09177. In the Introduction, page 2, the authors recall the question of Bart and Tsekanovskii asking whether the Banach-space operator relations equivalence after extension (EAE), matricial coupling (MC), and Schur coupling (SC) coincide. Since EAE and MC were known to coincide and SC was known to imply them, the remaining issue was whether

$$\text{EAE/MC} \Longrightarrow \text{SC}.$$

The same page says this matter had not been settled, then states the source paper's positive special case as Theorem 1.1: for inessential operators

$$\text{EAE} \Longleftrightarrow \text{EAOE} \Longleftrightarrow \text{SC},$$

with finite-dimensional extensions.

Supporting Answer

The later supporting paper is S. ter Horst, M. Messerschmidt, A. C. M. Ran, and M. Roelands, *Equivalence after extension and Schur coupling do not coincide, on essentially incomparable Banach spaces*, arXiv:1901.09254. Its abstract, page 1, explicitly identifies the 1994 Bart–Tsekanovskii question and states that the general answer is negative. It gives a concrete family: the forward shift operators on ℓ^p and ℓ^q, for $1 \leq p, q \leq \infty$ and $p \neq q$, are EAE but not SC.

The formal location is Theorem 2.1 on page 3. For operators on essentially incomparable Banach spaces, EAE is characterized by matching kernel and cokernel dimensions for Fredholm operators, while SC additionally requires Fredholm index zero. Thus operators with the same nonzero index can be EAE without being SC. The theorem states in particular that SC and EAE do not coincide.

Conclusion and Scope

The broad EAE/MC $\Rightarrow$ SC question recorded as unsettled in arXiv:1706.09177 is already answered by arXiv:1901.09254, and the answer is no in general. The source paper's own Theorem 1.1 remains an affirmative special case for inessential operators. This status note does not address later refined questions about Fredholm, relatively regular, or index-set variants; it only clears this 2017 broad-question target from the fresh-paper queue.

Local Evidence

The packet includes a local copy of the source PDF as `source_paper.pdf` and the decisive later paper as `supporting_paper_1901.09254.pdf`. The cheap run indexes were searched for the source arXiv id, title keywords, Schur coupling, inessential operators, and the supporting arXiv id. No prior solution packet for arXiv:1706.09177 was found; the only relevant memory hit was an attempt row for arXiv:1901.09254 itself.

References

[1] S. ter Horst, M. Messerschmidt, A. C. M. Ran, M. Roelands, and M. Wortel, *Equivalence after extension and Schur coupling coincide for inessential operators*, arXiv:1706.09177.

[2] S. ter Horst, M. Messerschmidt, A. C. M. Ran, and M. Roelands, *Equivalence after extension and Schur coupling do not coincide, on essentially incomparable Banach spaces*, arXiv:1901.09254.